

Multiple Choice (20 marks)

1. Elements with partially filled 4f or 5f orbitals include all of the following EXCEPT
 - (a) Cu
 - (b) Gd
 - (c) Eu
 - (d) Am
2. Helium effuses through a pinhole 5.33 times faster than an unknown gas. That gas is most likely
 - (a) CO₂
 - (b) CH₄
 - (c) C₅H₁₂
 - (d) C₈H₁₈
3. The number of allowed energy levels for the ⁵⁵Mn nuclide are:
 - (a) 3
 - (b) 5
 - (c) 8
 - (d) 4
4. The Pauli exclusion principle states that
 - (a) no two electrons can have the same four quantum numbers
 - (b) no two electrons with the same spin can occupy an orbital
 - (c) no two electrons can occupy separate orbitals
 - (d) no two electrons can pair up if there is an empty orbital available
5. Of the following solutions, which will have the highest ionic strength? (Assume complete dissociation.)
 - (a) 0.050 M AlCl₃
 - (b) 0.100 M NaCl
 - (c) 0.050 M CaCl₂
 - (d) 0.100 M HCl
6. Which of the following lists the electromagnetic spectral regions in order of decreasing wavelength?

- (a) ultraviolet, visible, infrared, X-ray
 - (b) X-ray, visible, ultraviolet, infrared
 - (c) X-ray, ultraviolet, visible, infrared
 - (d) infrared, visible, ultraviolet, X-ray
7. Which of the following is lower for argon than for neon?
- (a) Melting point
 - (b) Boiling point
 - (c) Polarizability
 - (d) First ionization energy
8. The photoelectron spectrum of carbon has three equally sized peaks. What peak is at the lowest energy?
- (a) The 1s peak has the lowest energy.
 - (b) The 2s peak has the lowest energy.
 - (c) The 2p peak has the lowest energy.
 - (d) The 1p peak has the lowest energy.
9. Calculate the Larmor frequency for a proton in a magnetic field of 1 T.
- (a) 23.56 GHz
 - (b) 42.58 MHz
 - (c) 74.34 kHz
 - (d) 13.93 MHz
10. The solid-state structures of the principal allotropes of elemental boron are made up of which of the following structural units?
- (a) B12 icosahedra
 - (b) B8 cubes
 - (c) B6 octahedra
 - (d) B4 tetrahedra

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'When will K_p and K_c have the same numerical value?' is 'At absolute zero for all reactions'.

2. True or False: The correct answer to 'Which of the following is true for Br₂ at standard temperature and pressure?' is 'It is a red-brown volatile liquid.'.
3. True or False: The correct answer to 'Which of the following can form hydrogen bonds?' is 'CH₃NH₂'.
4. True or False: The correct answer to 'The T₂ of an NMR line is 15 ms. Ignoring other sources of linebroadening, calculate its linewidth.' is '21.2 Hz'.
5. True or False: The correct answer to 'Which of the following experimental observations were explained by Planck's quantum theory?' is 'Temperature dependence of reaction rates'.
6. True or False: The correct answer to '0.0025 mol of a weak, monoprotic acid is dissolved in 0.250 L of distilled water. The pH was then measured as 4.26. What is the pK_a of...' is '7.92'.
7. True or False: The correct answer to 'Which of the following is expected to be the least soluble in water?' is 'CrCl₃'.
8. True or False: The correct answer to 'Which of the following statements about the lanthanide elements is NOT true?' is 'The most common oxidation state for the lanthanide elements is +3.'.
9. True or False: The correct answer to 'To prepare a buffer, all of the following are needed except' is 'triple-distilled water'.
10. True or False: The correct answer to 'What is the conjugate acid of the H₂PO₄⁻ ion?' is 'PO₄³⁻'.

End of Paper